Business Statistics Formula Sheet

Oct 22, 2026 · @Mats

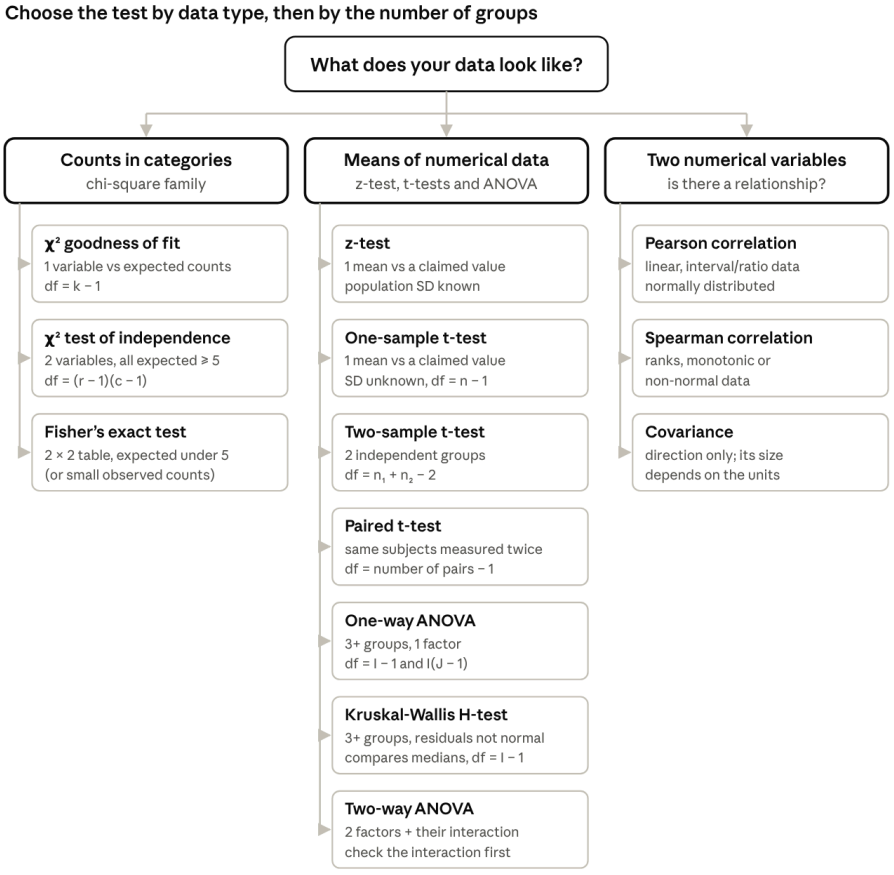
📄 Only the formulas on one page (with signs and when to use): Formula overview

Quick navigator: which formula when

Read the question: what does it **give** you, and what does it **ask** for? Find that row and jump to the section.

The question gives you	It asks for	Use	Section
Raw data (a list of numbers)	The average or the middle value	$\bar{x} = \Sigma x / n$, median (sort, take the middle)	0.4
Raw data	A percentile, Q1 or Q3	$i = (p / 100) \times n$	0.4
Raw data	How spread out it is (SD, variance)	$s = \sqrt{(\Sigma(x - \bar{x})^2 / (n - 1))}$	0.5
One value plus the mean and SD	How unusual it is, or comparing two values	$z = (x - \bar{x}) / s$	0.7
z plus the mean and SD	The value itself	$x = \bar{x} + z \cdot s$	0.7
Equally likely outcomes (die, coin, random pick)	P(A)	Outcomes in A / all outcomes	0.8
P(A)	P(not A), or "at least one"	$1 - P(A)$, $1 - P(\text{none})$	0.8
P(A), P(B) and P(A and B)	P(A or B)	$P(A) + P(B) - P(A \cap B)$	0.8
An average number per interval of time or space (λ)	P(exactly, at most or more than x events)	Poisson	0.9
n tries, the same p each try	P(exactly k successes)	Binomial	0.9
A range (a, b) where every value is equally likely	$P(c < X < d)$	Uniform: $(d - c) / (b - a)$	0.9
Two numerical variables (pairs of x and y)	Direction of the relationship	Covariance s_{xy}	2.1
Pairs of x and y, or s_{xy}, s_x and s_y	Direction and strength	Pearson $r = s_{xy} / (s_x \cdot s_y)$	2.2
Ranks, ordinal data, outliers, not normal	Strength of the relationship	Spearman ρ	2.2
μ, σ, n and a range	P(the sample mean falls in the range)	$z = (\text{boundary} - \mu) / (\sigma / \sqrt{n})$	3.1
μ, σ and one item (no n)	P(one item falls in the range)	$z = (x - \mu) / \sigma$	3.1
$\bar{x}$, σ, n and a confidence level	Confidence interval or margin of error	$\bar{x} \pm z \cdot \sigma / \sqrt{n}$	3.2
$\bar{x}$, a claimed μ_0, σ known, n	Test the claim	z-test: $z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$	4
Also N, and $n / N > 5\%$	The standard error	$SE \times \sqrt{((N - n) / (N - 1))}$	4
A test decision	Type I or Type II error (and its probability)	Templates, Type I = α	4.2 / 4.4

The question gives you	It asks for	Use	Section
Counts per category plus a predicted ratio	Does the data fit?	χ^2 goodness of fit, E = n × p, df = k − 1	5.1
A contingency table of two categorical variables	Are they related?	χ^2 independence, E = row × column / n	5.1
A 2×2 table with small counts (E < 5)	Are they related?	Fisher's exact test	5.2
$\bar{x}$, a claimed μ_0, s (σ unknown), n	Test the claim	One-sample t-test, df = n − 1	5.3
Two separate groups (means)	Are the means different?	Two-sample t-test, df = $n_1 + n_2 - 2$	5.3
The same subjects measured twice	Did it change?	Paired t-test, df = pairs − 1	5.3
3 or more groups, one factor	Are the means different?	One-way ANOVA F	6.1
Two factors	Effect of A, of B, and of the interaction	Two-way ANOVA (3 F tests)	6.2
Bartlett or Levene rejects (unequal variances)	Are the groups different?	Kruskal-Wallis	6.3
ANOVA rejected H_0	Which pair differs?	Bonferroni, α / k	6.3



Which test do I use? · 3 data types, 13 tests

Which test do I use? Read the diagram top-down: pick the column that matches your data, then the box that matches your number of groups and conditions.

Chapter 0 · Pre-course basics

0.1 Basic definitions

- **Population** = everyone (N, μ, σ). **Sample** = a part of it (n, x̄, s).
- **Categorical** = non-numerical. **Numerical**: discrete (separate points, e.g. counts) or continuous (any value in an interval).
- Cross-sectional = one moment in time. Time series = over several periods.
- **Descriptive** = summarising data. **Inferential** = using a sample to say something about the population.

0.2 Unethical behaviour

- Improper sampling
- Wrong analysis of the data
- Misleading graphs
- Wrong summary statistics
- Biased interpretation of results

0.3 Frequency

Frequency table = how often each value occurs. Relative frequency = frequency / n.

0.4 Measures of location

- **Mean:** $\bar{x} = \Sigma x_i / n$ (sample), $\mu = \Sigma x_i / N$ (population). Sensitive to outliers.
- **Median:** sort first. Odd n → the middle value. Even n → the average of the two middle values. Not sensitive to outliers.
- **Mode:** most frequent value. Two modes = bimodal, more = multimodal.
- **p-th percentile:** sort, then $i = (p/100) \times n$. Not a whole number → round **up** (8.4 → 9th value). ⚠️ My trap: whole number → the **average** of the i-th and (i+1)-th values.
- Quartiles: Q1 = 25th percentile, Q2 = median = 50th, Q3 = 75th.

$$\bar{x} = \frac{\sum x_i}{n} \qquad \mu = \frac{\sum x_i}{N} \qquad i = \frac{p}{100} \times n$$

🎯 **When to use:** x̄ when the data is a sample (most questions), μ only when it is the whole population. i gives the position of the p-th percentile in the sorted data (also for the median = 50th and Q1/Q3 = 25th/75th).

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0.5 Measures of variability

- Range = largest − smallest. IQR = Q3 − Q1.
- Standard deviation (the part under the root is the variance s^2 or σ^2; only use that on its own when a formula asks for it):

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} \qquad \sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

🎯 **When to use:** whenever a question asks how spread out the data is. s (divide by n − 1) for a sample, σ (divide by N) only when the data is the whole population.

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- The SD is in the same units as the data (USD 166), so report the SD for "how spread out".
- Steps: deviations ($x_i - \bar{x}$), which always add up to 0 (check!) → square → add up → divide → root.

0.6 Histogram

Bars per interval (class); height = frequency or relative frequency; the bars touch.

0.7 z-score

$$z_i = \frac{x_i - \bar{x}}{s} \qquad x_i = \bar{x} + z_i \cdot s$$

🎯 **When to use:** to say how many SDs one value is above or below the mean, or to compare values from two different data sets. The second form turns a z back into a value.

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- z = the number of SDs x_i is from the mean. z > 0 above the mean, z < 0 below, z = 0 on it.
- Same z in two data sets = same relative position. Use it to compare across groups (Finance z = 1.5 beats Marketing z = 0.5).

0.8 Probability

- **Sample space Ω** = the set of all possible outcomes. Coin once: Ω = {H, T}. 3 coins: $2^3 = 8$ outcomes {HHH, HHT, …, TTT}.
- Finite (countable end), infinite discrete (1, 2, 3, … no end), continuous (any value, e.g. rainfall ⩾ 20 mm).

- **Event A** = a subset of Ω (the outcomes you're interested in). **Complement A^c** = everything in Ω that is not in A.
- **Union A ∪ B** = A **or** B (or both). **Intersection A ∩ B** = A **and** B.
- **Disjoint** (mutually exclusive): A ∩ B is empty, the circles don't overlap.

$$P(A) = \frac{\text{number of ways A can occur}}{\text{total number of outcomes}}$$

🎯 **When to use:** when every outcome is equally likely (a fair die, a coin, a random pick): count the outcomes in A and divide by all outcomes in Ω.

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$$P(\Omega) = 1 \qquad 0 \le P(A) \le 1 \qquad P(A^c) = 1 - P(A)$$

🎯 **When to use:** $P(A^c)$ when "not A" is easier to count than A itself. "At least one" = 1 − P(none).

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$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad \text{disjoint: } P(A \cup B) = P(A) + P(B)$$

🎯 **When to use:** for "A or B" (either one, or both). Subtract the overlap when A and B can happen together. The short form only when they can't (disjoint).

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- ⚠️ Subtract the overlap once, otherwise you count it twice. "At least one" → often easier as 1 − P(none).
- Example: coin twice, Ω = {HH, HT, TH, TT}. A_1 = head first {HH, HT}, A_2 = head second {HH, TH}. $P(A_1 \cup A_2) = 0.5 + 0.5 - 0.25 = 0.75$.
- **Random variable** (capital X) = a number that depends on the outcome. Discrete = counting (items bought, customers). Continuous = measuring (time, weight, temperature).

🧠 **Mnemonic: "Discrete = Did I Count? Continuous = Can I Measure?"** Count it (emails, heads, customers) → discrete. Measure it with a ruler, clock or scale (time, weight, temperature) → continuous. Quick check: can 2.5 make sense? 2.5 emails no → discrete; 2.5 minutes yes → continuous.

Discrete (count it)	Continuous (measure it)
Number of customers in a shop	Time a customer spends in the shop
Number of children in a family	Height or weight of a child
Number of goals in a match	Distance a player runs
Number of cars in a car park	Speed of a car
Number of defective products in a box	Weight of a product
Number of likes on a post	Time spent scrolling
Number of rainy days in a month	Amount of rain (mm)
Number of employees	Salary per hour, temperature, travel time
⚠️ Shoe size (only fixed sizes: 42, 42.5)	⚠️ Age (time since birth, even when written in whole years)
—	⚠️ Money (usually treated as continuous: very many small steps)

Worked example (one die): Ω = {1, 2, 3, 4, 5, 6}. A = even = {2, 4, 6}, B = more than 4 = {5, 6}.

- A ∪ B = {2, 4, 5, 6} (each number once). A ∩ B = {6}.
- Counting:

$$P(A \cup B) = \frac{\text{number of ways } A \cup B \text{ can occur}}{\text{total number of outcomes}} = \frac{4}{6} = 0.6667$$

- Formula:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} = 0.6667$$

- Complement: A^c = {1, 3, 5}.

$$P(A^c) = 1 - P(A) = 1 - \frac{3}{6} = 0.5$$

- ⚠️ My traps: Ω is the **set** of outcomes, not "1" (1 = P(Ω)). In the formula use **how many** outcomes (B has 2), not the numbers themselves (5, 6), and divide by the total. A probability above 1 = a mistake.

0.8 Extra: conditional probability and independence

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad \text{independent: } P(A \cap B) = P(A) \cdot P(B)$$

🎯 **When to use:** "given that", "if we know that …" → conditional. Two independent events that both happen → multiply. Independent means P(A | B) = P(A).

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- Venn method: fill in the overlap first, then "A only" = P(A) − overlap, then "neither" = 1 − P(A ∪ B).
- ⚠️ Wording: "fewer than 10" = at most 9; "at most 10" includes 10. Two dice = 36 ordered outcomes: (1, 6) and (6, 1) are different.

0.9 Probability distributions

0.9.1 Discrete distributions

- Given as a table, a histogram or a formula. Every P(x) is between 0 and 1, and together they add up to 1.
- Example (children per household): P(X < 5) = 1 − P(X = 5) = 1 − 0.05 = 0.95.

Poisson distribution

- Counts how many times something happens in one interval of time or space: customers per 5 minutes, failures per week, typos per page.
- Assumptions: the events can be counted, they are independent, the average rate is constant, and two events never happen at exactly the same moment.

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \qquad x! = x \cdot (x-1) \cdots 1 \quad (0! = 1)$$

- **λ (lambda)** = the average number of events per interval. It is a count, not a probability (λ = 2: "on average 2 customers per 5 minutes"). x = the number of events you ask about. The formula gives the probability of exactly x events in one interval.
- ⚠️ λ must match the interval in the question: 2 per 5 minutes = 4 per 10 minutes.
- ⚠️ My trap: don't swap λ and x. λ is the **average** the question gives you ("on average 3 calls per minute" → λ = 3). x is the number you are **asked** about ("exactly 2 calls" → x = 2). λ sits next to "average", and x is in the question part.
- ⚠️ My trap: a longer interval changes λ, not x. "4 calls in 2 minutes" with 3 per minute → λ = 6, x = 4 (not "2 calls in 1 minute").
- ⚠️ My trap: for x = 0, $\lambda^0 = 1$ and 0! = 1, so **P(0) = e^(−λ)**. My calculator gives an error on 0!, so type only e^(−λ). Don't use λ^1 by mistake (then P(0) comes out the same as P(1)).

Special case x = 0 (no events):

$$P(X = 0) = \frac{e^{-\lambda} \cdot \lambda^{0}}{0!} = \frac{e^{-\lambda} \cdot 1}{1} = e^{-\lambda} \qquad \text{e.g. } \lambda = 0.5: \; e^{-0.5} = 0.6065$$

🎯 **When to use:** the question gives an average number per interval of time or space and asks the probability of exactly, at most or more than x events.

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- Example (λ = 2 customers per 5 min): P(0) = 0.1353, P(1) = 0.2707, P(2) = 0.2707, so P(X > 2) = 1 − (0.1353 + 0.2707 + 0.2707) = 0.3233.
- Not Poisson: a customer who waits until a cashier is free (the events are not independent).
- ⚠️ My trap: "more than 2" = 1 − P(X ⩽ 2) = 1 − [P(0) + P(1) + P(2)], not 1 − P(2).

Binomial distribution (used in the 5.1 dice example)

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- ⚠️ My trap: Poisson or binomial? Ask "is there a fixed n?" Yes (10 coin flips, 3 dice) → binomial, max = n. No, just an average per interval (typos per page, calls per minute) → Poisson. Measured (time, weight) → continuous, neither.

🎯 **When to use:** a fixed number of tries n, each one success or failure with the same probability p (number of sixes when you roll 3 dice: n = 3, p = 1/6). Poisson has no fixed n.

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0.9.2 Continuous distributions

- Density f(x) ⩾ 0 and the total area under the curve is 1. A probability is an area under the curve, so P(X = one exact value) = 0.

Uniform distribution

$$f(x) = \frac{1}{b-a} \quad (a < x < b) \qquad P(c < X < d) = \frac{d-c}{b-a}$$

🎯 **When to use:** every value between a and b is equally likely (a flat density, like a waiting time between 4 and 6 minutes).

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- Example (f = 0.5 on (4, 6)): P(4.5 < X < 5.5) = 1/2 = 0.5 and P(X > 5.5) = 0.5/2 = 0.25.
- Discrete uniform: a fair die, every outcome 1/6.

Chapter 1 · Continuous distributions and expectation

1.1 Continuous distributions

- **Normal** X ~ N(μ, σ²): the bell shape. μ = where the centre is, σ = how wide it is. A smaller σ makes it taller and narrower, and the area is always 1. You don't need the density formula itself.
- **Standard normal** Z ~ N(0, 1): μ = 0, σ = 1. Convert any normal X to Z, then use the Z-table.

$$Z = \frac{X - \mu}{\sigma} \qquad X = \mu + Z \cdot \sigma$$

🎯 **When to use:** X is normally distributed with a known μ and σ and the question asks a probability (Z-table), or gives a % and asks for the value (second form).

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- **χ^2_k (chi-squared):** skewed to the right, never negative. k = df, which is also the mean. Used for tests (chapter 5), not to describe real data. The bigger k is, the more it looks normal.
- **t-distribution:** a bell with fatter tails. Used when σ is unknown (only s), df = n − 1. From about 30 df it's almost Z. Used in the one-sample, two-sample and paired t-tests (5.3).
- **F-distribution:** the ratio of two χ^2's, each divided by its df. Compares variances (ANOVA, chapter 6).

$$F = \frac{Y_1/n_1}{Y_2/n_2}$$

🎯 **When to use:** comparing two variances. In practice you get F from ANOVA (chapter 6) and compare it with the F table.

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- **df (degrees of freedom)** = how many values can still vary freely (3 numbers with a fixed mean: 2 are free). df picks which t, χ^2 or F curve, so which row of the table. You get it from each test's rule, you never derive it.
- ⚠️ My trap: σ vs s. σ (Greek) = SD of the whole population, given in the question → **Z**. s (Latin) = SD of my sample, calculated from the data → **t**, df = n − 1. "σ unknown" or "sample SD" → t.
- ⚠️ My trap: "Which distribution / which test / how many df?" means **name** it, don't calculate. Example: 10 bags, σ unknown → t-distribution, df = 9.
- ⚠️ My trap: one tail or two? "Heaviest 5%" = one tail → 1.645, not 1.96 (see the mnemonic in 3.2).

1.2 Expectation, variance and CDF

- **E(X)** = the long-run average if you repeat the experiment many times. It doesn't have to be a possible value.

$$E(X) = \mu = \sum x_i \, p(x_i)$$

🎯 **When to use:** a discrete distribution (table of values with probabilities) and the question asks for the expected value, the average outcome, or the expected gain of a game.

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- Die: E(X) = 1·⅙ + 2·⅙ + … + 6·⅙ = 3.5. Game (even → win $2, odd → lose $1): E = 2·½ − 1·½ = $0.50.

$$Var(X) = \sigma^2 = \sum (x_i - \mu)^2 \, p(x_i) \qquad \sigma = \sqrt{Var(X)}$$

🎯 **When to use:** the spread of a discrete distribution given as a table. Like 0.5, but each squared deviation is weighted by $p(x_i)$ instead of dividing by n.

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- Shortcuts: Poisson E(X) = Var(X) = λ. Normal E(X) = μ, Var(X) = σ². Continuous: the same with an integral (not calculated on the exam).

$$F(x_0) = P(X \leq x_0) = \sum_{x_i \leq x_0} p(x_i)$$

🎯 **When to use:** "at most" or "less than" probabilities: the CDF is the area to the left of x_0. The Z-table is the CDF of Z. Discrete version: the cumulative relative frequency (add up row by row).

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- Z-table gives the area to the **left** of z. Birth weight (μ = 7, σ = 1): P(6.5 ⩽ X ⩽ 8) = P(−0.5 ⩽ Z ⩽ 1) = 0.8413 − 0.3085 = 0.53.

1.3 Extra: distribution facts and shortcuts

- ⚠️ N(μ, σ²): the second number is the **variance**. N(100, 225) → σ = √225 = 15.
- Empirical rule (normal): about 68% within 1σ of the mean, 95% within 2σ, 99.7% within 3σ.

$$\text{Binomial}: \quad E(X) = np, \quad \text{Var}(X) = np(1-p)$$

$$\text{Uniform}: \quad E(X) = (a+b)/2, \quad \text{Var}(X) = (b-a)^2/12$$

🎯 **When to use:** the expected value or variance of a binomial (n tries, p) or uniform (a, b) variable, without a table.

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$$Var(X) = E(X^2) - [E(X)]^2 \qquad E(X^2) = \sum x_i^2 \, p(x_i)$$

🎯 **When to use:** a quicker way to get Var(X) from a probability table. Square first, then weight ($E(X^2)$); that is not the same as $[E(X)]^2$.

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$$E(aX + b) = a\,E(X) + b \qquad Var(aX + b) = a^2\,Var(X)$$

🎯 **When to use:** every value is multiplied by a and/or shifted by b (prices +10%, a fixed fee). Shifting (+ b) never changes the spread.

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$$P(X > x) = 1 - F(x) \qquad P(a < X \leq b) = F(b) - F(a)$$

🎯 **When to use:** "more than" or "between" with a CDF or the Z-table (which only gives the area to the left).

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Chapter 2 · Covariance and correlation

2.1 Covariance (direction only)

- Sample: $s_{xy} = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$. Population: $\text{cov}(X, Y) = \Sigma(X_i - \bar{X})(Y_i - \bar{Y}) / N$.
- Positive = both go up together; negative = one up, other down.
- Both above or both below their mean (quadrants I and III) → positive product; one above, one below (II and IV) → negative.
- ⚠️ Any number, depends on the units, so it gives the **direction but not the strength**.
- Covariance matrix: variances on the diagonal, covariances off it; symmetric because cov(x, y) = cov(y, x).

$$s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n - 1} \qquad \sigma_{xy} = \frac{\sum(x_i - \mu_x)(y_i - \mu_y)}{N}$$

🎯 **When to use:** two numerical variables, when you only need the direction (positive or negative) of the linear relationship, or as the step towards r. s_xy for a sample, σ_xy for a population.

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2.2 Correlation (direction and strength)

Pearson: $r_{xy} = s_{xy} / (s_x \cdot s_y)$. Dividing by both SDs removes the units ("not affected by the units of measurement"), so $-1 \leq r \leq 1$. Population: ρ_{xy}.

$$r_{xy} = \frac{s_{xy}}{s_x\, s_y} \qquad \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x\, \sigma_y} \qquad -1 \leq r \leq 1$$

🎯 **When to use:** direction and strength of a linear relationship (always between −1 and 1). Interval/ratio data, linear, roughly normal.

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Value of r (ignore the sign)	Strength
.00 – .19	very weak
.20 – .39	weak
.40 – .59	moderate
.60 – .79	strong
.80 – 1.0	very strong

Use Pearson only when: interval/ratio data, **linear** relationship, normally distributed.

Spearman rank: $\rho = 1 - 6\Sigma d_i^2 / (n(n^2 - 1))$, with d_i = rank of x − rank of y.

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \qquad d_i = R_x - R_y$$

🎯 **When to use:** ranked/ordinal data, data that isn't normal, outliers, or a curve that keeps going the same way (monotonic). Rank both variables first.

✍️ **Best practice:** …

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1. Rank x (1 = lowest, or highest; same direction for both). 2. Rank y. Ties (two values the same): give both the middle (average) rank, then continue counting. E.g. two values that would get places 2 and 3 both get (2 + 3)/2 = 2.5, and the next value gets rank 4. Three values on places 2, 3, 4 all get 3, and the next gets 5. 3. d = $R_x - R_y$. 4. Square, add up. 5. Fill in.

Use Spearman when: ordinal/ranked data, not normal, or a curve that keeps going the same way (monotonic). Example $y = e^x$: Pearson 0.70, Spearman 1.

⚠️ Pearson traps: small sample, outliers, effect too small to be useful, non-linear relationships missed (a curve can give r ≈ 0).

2.1–2.2 How to calculate and what it means

Calculating (table method)

1. Means $\bar{x}$ and $\bar{y}$. 2. Columns $x - \bar{x}$ and $y - \bar{y}$ (each column always adds up to 0: check!). 3. **Multiply per row**, then add up the products. ⚠️ My trap: adding the columns first, then multiplying. 4. Divide by n − 1 (sample) or N (population). 5. For s_x and s_Y: columns $(x - \bar{x})^2$ and $(y - \bar{y})^2$, add up, divide, take the root.

Write it out: $s_{xY} = [(-1)(-1) + (2)(1) + (-2)(-2) + (1)(2) + (0)(0)] / 4 = 9/4 = 2.25$.

Sample or population? Population (N) only when the question says it is everyone ("all employees"); otherwise sample (n − 1). State your choice, and use the same divisor for the covariance and the SDs (r comes out the same either way).

What the answer means (exam sentences)

- Covariance: only the sign counts. "The covariance is positive (1.8), so there is a positive linear relationship between hours studied and grade. The value itself says nothing about the strength."
- Pearson r: sign = direction, size = strength (table above). "r = 0.90: a very strong positive linear relationship between hours studied and grade."
- Spearman ρ: same reading, but about ranks / a monotonic relationship. "ρ = 0.90: a very strong positive monotonic relationship: students who rank higher in hours also rank higher in grade."

Which one when

Measure	Tells you	Use when
Covariance s_{xy}	Direction only (any number, depends on units)	Step towards r; covariance matrix
Pearson r	Direction + strength of a **linear** relationship (−1 to 1)	Interval/ratio data, straight-line pattern, roughly normal
Spearman ρ	Direction + strength of a **monotonic** relationship (−1 to 1)	Ranked/ordinal data, not normal, outliers, or a curve that keeps going one way

Chapter 3 · Standard error and confidence intervals

3.1 Probability that a sample mean falls in a range

Use this when a question asks how likely the average of a sample (n given) is to land between two values.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \qquad P = \text{table}(z_{\text{upper}}) - \text{table}(z_{\text{lower}})$$

🎯 **When to use:** the probability that the mean of a sample (n given, σ known) is above, below or between values.

✍️ **Best practice:** …

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The trick: you don't have x̄. Put each boundary of the range into the formula where x̄ would go, one at a time.

1. Standard error = σ / √n
2. z for the upper boundary and z for the lower boundary
3. Look both up in the Z-table (it gives the area to the left)
4. P = table(upper z) − table(lower z)

Example: μ = 200 g, σ = 20 g, n = 64. P(196 ⩽ x̄ ⩽ 204)? Standard error = 20 / √64 = 2.5, so z = (204 − 200) / 2.5 = 1.6 and z = (196 − 200) / 2.5 = −1.6. Then P = table(1.6) − table(−1.6).

One item instead of a sample average (no n, "a randomly chosen…")? Divide by σ instead: z = (x − μ) / σ.

$$SE = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \qquad \text{one item}: \; z = \frac{x - \mu}{\sigma}$$

🎯 **When to use:** SE whenever you work with a sample mean. z = (x − μ)/σ (no √n) for one single item ("a randomly chosen bag").

✍️ **Best practice:** …

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3.2 Confidence intervals, and which z to use (also 4.3)

Confidence interval: x̄ ± z · σ/√n. Margin of error = z · σ/√n. Pick z from the confidence level (two-sided column below).

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \qquad ME = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

🎯 **When to use:** estimating μ with a range at a confidence level (90%, 95%, 99%) when σ is known. ME when the question asks for the margin of error (precision).

✍️ **Best practice:** …

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The sign in H_a points at the tail where α goes: > right, < left, ≠ both (α/2 each).

Confidence / α	Two-sided (≠) and CI	Right-sided (>)	Left-sided (<)
90% / α = 0.10	±1.645 (0.05 each tail)	1.28	−1.28
95% / α = 0.05	±1.96 (0.025 each tail)	1.645	−1.645
99% / α = 0.01	±2.576 (0.005 each tail)	2.33	−2.33

Only halve α when the test is two-sided. Reject H_0 if p-value ⩽ α (two-sided in the course notes: compare the tail area with α/2).

Finding it in the Z-table (left areas): right-sided, look up 1 − α; left-sided, look up α; two-sided, look up 1 − α/2.

🧠 Mnemonic: **"One tail, one-six-four-five. Two tails, nineteen-sixty. Ninety-nine, two-five-seven-five."**

- 5% in **one** tail (> or <, "the heaviest 5%") → z = **1.645**
- 5% split over **two** tails (≠, 95% CI) → z = **1.96**
- 1% split over **two** tails (99% CI) → z = **2.575**

$$x = \mu + z \cdot \sigma \qquad (\text{top } 5\%: \ z = 1.645; \ \text{lowest } 10\%: \ z = -1.28)$$

🎯 **When to use:** the question gives a percentage (top 5%, lowest 10%) and asks for the boundary value (a weight, a score).

✍️ **Best practice:** …

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3.3 Extra: sampling distribution and more intervals

- SD = spread of the individual values. SE = spread of the sample means. The SE is always smaller (n > 1). 4× n halves the SE.

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \qquad SE = \frac{\sigma}{\sqrt{n}} \qquad z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

🎯 **When to use:** central limit theorem. A question about a sample mean when the population is not normal (or unknown): with n ≥ 30 (n ≥ 50 if very skewed) you may still use z. Population normal → any n.

✍️ **Best practice:** …

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$$\bar{x} \pm t_{\alpha/2,\ n-1} \cdot \frac{s}{\sqrt{n}} \qquad df = n - 1$$

🎯 **When to use:** a confidence interval when σ is unknown (only s) and the sample is small. t from the t table, two-tailed column, df = n − 1.

✍️ **Best practice:** …

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$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad \hat{p} = \frac{\text{successes}}{n}$$

🎯 **When to use:** a proportion or percentage ("the share of customers who…", p̂ = successes / n). Always z.

✍️ **Best practice:** …

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$$\bar{x} = \frac{A+B}{2} \qquad ME = \frac{B-A}{2}$$

🎯 **When to use:** the interval [A, B] is given and the question asks for x̄ or the margin of error.

✍️ **Best practice:** …

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- **z or t?** σ known → z. σ unknown but n ≥ 30 → z is fine. σ unknown and n small → t (df = n − 1). Proportion → always z.
- **Width:** higher confidence or a larger σ → wider. Larger n → narrower (4× n halves the margin). t instead of z → slightly wider.
- **Interpretation:** "We are 95% confident that μ lies between A and B." ⚠️ Not "95% of the data" and not "a probability of 0.95 that μ is in this interval" (μ is fixed, the interval varies).

Chapter 4 · Hypothesis testing

4.1–4.3 Hypothesis test: step by step (with my traps)

Step 1. Hypotheses (in symbols, always μ, never x̄)

- **H_0 = the claim / status quo** (the company, or "nothing changed"). Usually =, ≤ or ≥, but my teacher says H_0 can also be < or > when that is the status quo claim (e.g. NS claims less than 6.5 h → H_0: μ < 6.5, H_a: μ ≥ 6.5).
- **H_a = what the researcher suspects** (the challenger): <, > or ≠.
- **The rule:** H_0 = the existing claim or situation, and H_a = what someone is trying to prove. Watch out for words like "reduces", "increases" and "more": they almost always belong in H_a.
- Direction words: more / longer / increased → >. Less / shorter / reduced → <. "Different" / "is the claim true?" / no direction → ≠.
- ⚠️ **My trap: I swap H_0 and H_a.** Check: is my H_0 the claim / status quo, not the suspicion? If not, swap.

Step 2. α and the tail(s)

- Ask: **is there a ≠ in H_a?** Yes → halve α (two tails). No → keep the full α in one tail.
- The sign in H_a points at the tail: > right, < left, ≠ both.
- ⚠️ **My trap: halving α in a one-sided test.** Only ≠ gets halved.

Step 3. Test statistic

- SE = σ/√n (if n/N > 5%, multiply by √((N−n)/(N−1))).
- z = (x̄ − μ_0) / SE, with μ_0 from H_0.
- ⚠️ Use brackets on the calculator: (x̄ − μ_0) ÷ SE.
- ⚠️ **My trap: filling in 1.96 because I see α = 0.05.** In a test I *calculate* z from the data. 1.96 / 1.645 are only boundaries (CI, or the edge of the rejection region), and 1.96 is only for two-sided.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \qquad SE_{\text{FPC}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

🎯 **When to use:** hypothesis test about one mean when σ is known (z-test). Use SE_FPC instead of σ/√n when the sample is more than 5% of the population (n/N > 0.05).

✍️ **Best practice:** …

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Step 4. p-value

- Meaning: *if H_0 were true, the probability of a sample at least this extreme.* Small p = weird sample = evidence against H_0.
- Left-sided (<): p = table(z). Right-sided (>): p = 1 − table(z). Two-sided (≠): compare the tail area with α/2.

Two-sided (≠): the Z-table gives only one tail, but the test counts both. Two ways, same answer:

- **Way 1 (my notes):** compare one tail with α/2. Example: 0.0228 < 0.025 → reject H_0.
- **Way 2:** p = 2 × one tail, compare with α. Example: 0.0456 < 0.05 → reject H_0.
- If the answer options show a doubled p-value (like 0.0456), they used Way 2.
- ⚠️ Only for ≠. For < or >, the table value (or 1 − table) IS the p-value: compare with the full α.

Step 5. Decision

- p ⩽ α → reject H_0 and accept H_a (allowed in my course). p > α → do not reject H_0. **"If p is low, H_0 must go."**
- ⚠️ Never write "accept H_0" or "H_0 is proven". Accepting H_a is fine, accepting H_0 is not.

	What it means	Decision	Conclusion sentence
p ⩽ α (p smaller)	Sample too weird to believe H_0	Reject H_0 (accept H_a)	"There is enough evidence that … [H_a]"
p > α (p bigger)	Nothing strange happened	Do not reject H_0	"There is not enough evidence that … [H_a]"

α is fixed beforehand; only p comes from the data. Always ask: is my p below the line α?

Step 6. Conclusion sentence

- "p = 0.0228 < α = 0.05, so we reject H_0. There is enough evidence that … [H_a in words]."
- "p = 0.62 > α = 0.05, so we do not reject H_0. There is not enough evidence that … [H_a in words]."

4.2 and 4.4 **Type I and Type II errors**

1. First write H_0 and H_a in words.
2. Fill in the templates:

- **Type I** (false alarm, probability α): *we conclude* [H_a], *but in reality* [H_0] is true.
- **Type II** (missed effect, probability β): *we conclude* [H_0, word for word], *but in reality* [H_a] is true.
- **Power** = 1 − β = the probability of correctly rejecting a false H_0 (catching the guilty).
- ⚠️ **My traps:** describing what is true instead of starting with "we conclude"; inventing a new opposite ("smaller" instead of "not taller"); labelling the error sentences H_0/H_a. The errors are Type I and Type II.

4.5 Extra: β, power and the link with confidence intervals

$$\bar{x}_c = \mu_0 \pm z_\alpha \frac{\sigma}{\sqrt{n}} \qquad \beta = P(\text{do not reject } H_0 \mid \mu = \mu_1) \qquad \text{power} = 1 - \beta$$

🎯 **When to use:** "the probability of a Type II error if the true mean is …", or "the power of the test".

✍️ **Best practice:** …

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- Lamp example (H_0: $\mu \geq 120$, H_a: $\mu < 120$, SE = 2, $\alpha = 0.05$): cut-off = $120 - 1.645 \times 2 = 116.71$. True $\mu = 112$: $\beta = P(\bar{x} > 116.71) = P(Z > 2.36) = 0.0091$, power = 0.9909.
- True μ closer to μ_0 → higher β, lower power. Smaller α → more Type II errors. Larger n → lower β (the only way to reduce both errors).
- Link with CIs: μ_0 outside the 95% CI → reject H_0 in a two-sided test at $\alpha = 0.05$.

Chapter 5 · Chi-square tests

5.1 Chi-square goodness-of-fit test

Use this when you have **counts per category** and want to check whether they fit an expected distribution (e.g. "is the die fair?").

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

🎯 **When to use:** counts per category (categorical data). The same χ^2 formula works for goodness of fit and for the test of independence; only E and df differ.

✍️ **Best practice:** …

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O = observed count, E = expected count if H_0 is true.

1. H_0: the data follow the distribution (e.g. the dice are fair). H_a: they don't.
2. Expected counts: E = total n × probability per category (e.g. a binomial probability). Check that the E column adds up to n.
3. Table: O, E, $(O-E)^2$, $(O-E)^2/E$. Add up the last column → test statistic χ^2.
4. df = number of categories − 1.
5. Boundary χ^2 from the table below (row df, column α).
6. Test χ^2 **bigger** than the boundary → reject H_0 (accept H_a). **Smaller** → do not reject H_0.

$$E_i = n \cdot p_i \qquad df = k - 1$$

🎯 **When to use:** goodness of fit: one categorical variable, E from the predicted proportions p_i (fair die, a claimed ratio), df = categories − 1.

✍️ **Best practice:** …

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- Always a right tail: χ^2 is never negative, and only a big χ^2 means a bad fit. Don't halve α, don't do 1 − area.
- Bigger χ^2 = worse fit. $\chi^2 = 0$ = a perfect fit.
- Example (3 dice, 50 rolls): $\chi^2 = 0.86 + 0.09 + 3.5 + 2.25 = 6.7$, df = 4 − 1 = 3, boundary 7.815. 6.7 < 7.815, so we do not reject H_0: there is not enough evidence that the dice are unfair.
- ⚠️ **My trap:** the slide says "we accept our null hypothesis". On the exam, write "we do not reject H_0".
- **Requirements (check them first, my course asks this):** a simple random sample; categorical / nominal data (not continuous); at least 5 **expected** in every category. Not needed: a normal distribution.

χ^2 table (right-tail area α)

df	α = .995	α = .99	α = .975	α = .95	α = .90	α = .10	α = .05	α = .025	α = .01	α = .005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215

df	α = .995	α = .99	α = .975	α = .95	α = .90	α = .10	α = .05	α = .025	α = .01	α = .005
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

How to read it (Scribbr layout): α = the area to the RIGHT of the value. Goodness of fit and independence tests: use the normal α column (.10, .05, .01). The columns .90 to .995 are only for left-tailed or two-tailed variance tests (left tail = column 1 − α). df not in the table (e.g. 46)? Round down to the next row (40).

Other df: Python `chi2.ppf(1 - α, df)` (scipy.stats), Excel `=CHISQ.INV.RT(α; df)`.

5.1 Chi-square test of independence

Use this when you have **two categorical variables** in a contingency table and ask "are they related?" ("association between", "independent of").

1. H_0: variable 1 is independent of variable 2 (not related). H_a: they are not independent (related).
2. Check the requirements: random sample, two categorical variables, every **expected** cell ⩾ 5.
3. Add row totals, column totals and the grand total n.
4. Expected per cell: **E = row total × column total / n**.
5. Per cell $(O - E)^2/E$, add up all cells → χ^2.
6. **df = (rows − 1) × (columns − 1)**. Don't count the total row or column.
7. Boundary from the χ^2 table above. χ^2 > boundary → reject H_0 (accept H_a: there is a relationship).

$$E_{ij} = \frac{\text{row total} \times \text{column total}}{n} \qquad df = (r-1)(c-1)$$

🎯 **When to use:** test of independence: two categorical variables in a contingency table ("are they related?"). All E ⩾ 5, otherwise for a 2×2 table use Fisher.

✍️ **Best practice:** …

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- Example (education × married once/more, n = 1436): E college & once = 611 × 1231 / 1436 = 523.8. χ^2 = 1.31 + 7.87 + 0.97 + 5.83 = 15.98, df = (2 − 1)(2 − 1) = 1, boundary 3.841. 15.98 > 3.841, so we reject H_0: there is a relationship between education and marital status.
- Example (ticket class × flight length, n = 920): χ^2 = 100.43, df = (3 − 1)(2 − 1) = 2, boundary 5.991 → reject H_0.
- In a 2×2 table every (O − E) has the same size, only the sign flips (+26.2 / −26.2). Handy check.

5.2 Fisher's exact test

- Same question as the test of independence (is there an association?), but only for a **2×2 table**.

- Use it when the numbers are small: observed count about < 10, or expected count < 5. Then χ^2 is no longer accurate, because χ^2 is an approximation and Fisher is exact.
- Calculated by software, not by hand: Python `stats.fisher_exact(table)` (table must be 2×2), gives a p-value. $p \leq \alpha \rightarrow$ reject H_0 (there is an association).
- Bigger than 2×2 with small numbers: not in Python, use other software (SPSS).
- Example (Bower 2003, Letterman 8 right / 1 wrong vs Shaffer 4 / 5, n = 18): expected = 9 × 12 / 18 = 6 and 3, so two cells are < 5. $\chi^2 = 4.000$, $p = 0.046 < 0.05$ says reject H_0, **but** Fisher gives $p = 0.1312 > 0.05$: do not reject H_0. With small counts, trust Fisher; χ^2 can give the wrong conclusion.
- Why "exact": it uses the hypergeometric distribution (all tables with the same row and column totals) instead of the χ^2 approximation.
- Python output `(4.5, 0.115)` = (odds ratio, **p-value**). Compare the second number with α.

Which test when (chapter 5)

Situation	Test
One categorical variable, counts vs a predicted ratio or distribution	χ^2 goodness of fit, df = categories − 1
Two categorical variables, "related / associated / independent?"	χ^2 test of independence, df = (r − 1)(c − 1)
Same, but 2×2 with small counts (expected < 5)	Fisher's exact test
A mean compared with a claimed value	Not χ^2: one-sample t-test

5.3 t-test (comparing means)

For **continuous** data (means), at most **two groups**. More than two groups → not a t-test. Uses s (sample SD) instead of σ, so look up the boundary in the **t table** (bottom of this sheet) instead of the z table.

One-sample t-test: one sample mean vs a known / claimed value μ_0 (e.g. screws should be 20 mm, retirement age 65).

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \qquad df = n - 1$$

🎯 **When to use:** one sample mean vs a claimed value μ_0 when σ is unknown (you only have s). σ known → the z-test from 4.1–4.3.

✍️ **Best practice:** …

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- H_0: $\mu = \mu_0$. H_a: $\mu \neq \mu_0$ (or < / > if the question gives a direction).
- Requirements: random sample, continuous data, roughly normal (check a histogram: bell-shaped, no outliers).
- Same steps as chapter 4, only t instead of z: |t| > boundary t → reject H_0.

- Example (31 screws): $\bar{x}$ = 21.40, s = 2.54. t = (21.40 − 20) / (2.54/√31) = 3.07. df = 30, two-sided α 0.05 → boundary 2.042. 3.07 > 2.042 → reject H_0: the mean diameter is not 20 mm.

Two-sample (independent) t-test: the means of two **separate** groups (pizza company A vs B, production line 1 vs 2).

- H_0: the population means are equal ($\mu_1 = \mu_2$). H_a: they are not equal.
- Requirements: independent groups, random samples, roughly normal, **equal variances** (similar SDs), no outliers.
- Not calculated by hand in this course: Python `stats.ttest_ind(a, b)` → (statistic t, pvalue).
- **df = n_1 + n_2 − 2.**
- Example (screws, line 1 vs line 2): t = 0.524, p = 0.602. df = 31 + 31 − 2 = 60 → boundary 2.000. 0.524 < 2.000 (and p 0.602 > 0.05) → do not reject H_0: no evidence that the lines differ.
- ⚠️ **My trap:** the slide says "we accept our null hypothesis" again. On the exam: "we do not reject H_0".

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}} \qquad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

🎯 **When to use:** two-sample t-test: the means of two separate (independent) groups. df gives the row in the t table. Check equal variances with Levene first.

✍️ **Best practice:** …

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- Checking equal variances in Python: `stats.levene(a, b)`. Here p = 0.94 > 0.05, so the variances are equal → `ttest_ind(a, b, equal_var=True)`.

Paired (dependent) t-test: two measurements on the **same** subjects (blood pressure before and after a treatment).

- H_0: the mean difference between the pairs = 0. H_a: the mean difference ≠ 0.
- Requirements: continuous (interval/ratio), independent observations, differences roughly normal, no significant outliers.
- Python `stats.ttest_rel(before, after)`.

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \qquad d = \text{before} - \text{after} \qquad df = \text{pairs} - 1$$

🎯 **When to use:** paired t-test: two measurements on the same subjects (before/after). df = number of pairs − 1, not all measurements.

✍️ **Best practice:** …

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- Example (BP.csv, 120 patients): mean before 156.45, after 151.36. `ttest_rel` → t = 3.34, p = 0.0011. **df = number of pairs − 1 = 119.** p 0.0011 < 0.05 → reject H_0 and accept H_a: the mean blood pressure changed after the treatment.

Which t-test?

Situation	Test	Python
One group vs a claimed number	One-sample t-test, df = n − 1	`stats.ttest_1samp`
Two different groups	Two-sample (independent) t-test, df = $n_1 + n_2 - 2$	`stats.ttest_ind`
Same people measured twice (before/after)	Paired t-test	`stats.ttest_rel`
Counts in categories	Not a t-test: χ^2 (5.1) or Fisher (5.2)	

Chapter 6 · ANOVA

6 Why ANOVA

- Comparing the means of **more than two groups**. A t-test only works for two groups.
- Why not do many t-tests? (1) Too many tests: 10 groups = 10C2 = 45 pairs. (2) The Type I error adds up with every test.

6.1 One-way ANOVA

One factor (independent variable, e.g. website design) with I levels (groups), and one dependent variable (e.g. amount spent).

- H_0: $\mu_1 = \mu_2 = \ldots = \mu_I$ (all group means equal: the factor has no effect).
- H_a: at least two of the group means are not equal (the factor has an effect).
- Test statistic F = variance **between** the groups / variance **within** the groups. Always a right tail (one-sided).

Steps (course method)

1. Grand mean μ = the mean of all data points. Group means $\bar{y}_i$.
2. $\beta_i = \bar{y}_i - \mu$ for each group. Check: the β's add up to 0.
3. SSE per group = $\Sigma(y_j - \bar{y})^2$ (like a variance, but without dividing by n − 1). SSE total = $SSE_1 + SSE_2 + \ldots$
4. SSTR = $\beta_1^2 + \beta_2^2 + \ldots$ (between-groups variation).
5. F-ratio:

$$F = \frac{SSTR/(I-1)}{SSE/(I(J-1))} \qquad DF1 = I - 1, \quad DF2 = I(J-1)$$

$$\beta_i = \bar{y}_i - \mu \qquad SSE = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2 \qquad SSTR = \sum_i \beta_i^2$$

🎯 **When to use:** one-way ANOVA: the means of more than two groups with one factor. β, SSE and SSTR (the formulas above) are the steps towards F.

✍️ **Best practice:** …

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I = number of groups (treatments), J = number of data points per group. **DF1 = I − 1** (goes with SSTR, the columns of the F table). **DF2 = I(J − 1)** (goes with SSE, the rows).

6. Boundary from the F table (bottom of this sheet, α = 0.05). F > boundary → reject H_0: at least one group mean differs. F < boundary → do not reject H_0.

- If H_0 is true, F is close to 1. A big F (lots of variation between groups, little within) → reject H_0.
- Example (3 website designs, 7 customers each): group means 59.28, 110, 84.43; SSTR = 1288, SSE = 114683. F = (1288/2) / (114683/18) = 644/6371 = 0.10. DF1 = 2, DF2 = 3 × 6 = 18 → boundary 3.5546. 0.10 < 3.5546 → do not reject H_0: no evidence that website design affects spending.
- ⚠️ Note: textbooks and Python (`stats.f_oneway`) use SSTR = J × Σβ² (each β² counted once per data point). That gives F = 0.71, p = 0.51 for this example. Same conclusion. On the exam, use the course formula unless the question says otherwise.
- Data = fit + residual: ANOVA fits a linear model; the residuals (errors) are what the model can't explain. (More in the regression lecture.)

Limitations / requirements

- Only one factor and one dependent variable.
- Tells you **that** at least one pair of means differs, **not which** pair.
- Dependent variable roughly normal in every group, similar variance in every group, samples independent.

6.2 Two-way ANOVA

Two categorical factors (e.g. A = website design, B = gender) and one numerical dependent variable (spending).

- I = levels of factor A, J = levels of factor B, K = observations per combination. **N = I × J × K** (example: 3 × 2 × 4 = 24).
- Advantages: two questions in one test ("two one-way ANOVAs for the price of one"); shows **interaction** between the factors; **more power** (1 − β) than one-way ANOVA.
- SSE_total = SSE_A + SSE_B + SSE_AB + SSE_unexplained (within each of the I × J groups).

$$SS_{\text{total}} = SSE_A + SSE_B + SSE_{AB} + SSE_{\text{unexpl}} \qquad N = I \times J \times K$$

🎯 **When to use:** two-way ANOVA (two factors). N = all observations. The total variation splits into factor A, factor B, interaction and unexplained.

✍️ **Best practice:** …

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Three hypothesis tests at once

Test	H_0	F	df
Factor A (main effect)	All levels of A have the same mean (design has no effect)	SSE_A/(I − 1) ÷ SSE_unexpl/(N − IJ)	I − 1 and N − IJ
Factor B (main effect)	All levels of B have the same mean (gender has no effect)	SSE_B/(J − 1) ÷ SSE_unexpl/(N − IJ)	J − 1 and N − IJ
Interaction A×B	The effect of A does not depend on B (men and women don't react differently to the designs)	SSE_AB/((I − 1)(J − 1)) ÷ SSE_unexpl/(N − IJ)	(I − 1)(J − 1) and N − IJ

$$F_{effect} = \frac{SSE_{effect}/df_{effect}}{SSE_{\text{unexpl}}/(N-IJ)} \qquad df_A = I-1, \quad df_B = J-1, \quad df_{AB} = (I-1)(J-1)$$

🎯 **When to use:** two-way ANOVA: one F per H_0 (factor A, factor B, interaction). Look at F_AB first.

✍️ **Best practice:** …

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- Decision per test as usual: p ≤ α (or F > boundary) → reject that H_0.
- ⚠️ **Check the interaction first.** Interaction significant (p ≤ α) → do **not** interpret the main effects of A and B. Instead do separate one-way ANOVAs (e.g. one for men, one for women).
- H_0 always = "nothing new is happening / no relationship".

6.3 ANOVA assumptions (and what to do if they fail)

Assumption	How to check	If it fails
Independent samples	Good experiment design	—
Normality: residuals ~ normal (mean 0, variance σ^2)	Q-Q plot / histogram of the residuals	Use **Kruskal-Wallis** instead

Assumption	How to check	If it fails
Equal variance in every group	Boxplots; **Bartlett** (normal data) or **Levene** (non-normal data). H_0: all variances are equal	$p \leq 0.05$ → variances differ → use Kruskal-Wallis

Kruskal-Wallis H-test (non-parametric ANOVA)

- H_0: the population **medians** of all groups are equal. H_a: at least one differs.
- H statistic follows a χ^2 distribution with **df = I − 1** → one-tailed, use the χ^2 table (5.1).
- 2 or more independent groups, sizes may differ, at least 5 measurements per group.
- Python: `stats.kruskal(g1, g2, g3, …)`.

Post hoc: Bonferroni (which pair differs?)

- ANOVA only says *that* a difference exists. Bonferroni does pairwise t-tests, each at **α / k** (k = number of pairs), so the overall Type I error (FWER, family-wise error rate) stays α.
- Example: 4 groups → k = 4C2 = 6 pairs → 0.05 / 6 = 0.008. A pair differs if its p < 0.008 (or corrected p < 0.05).

$$\alpha^* = \frac{\alpha}{k} \qquad k = \mathrm{C}(I, 2) = \frac{I(I-1)}{2}$$

🎯 **When to use:** after ANOVA (or Kruskal-Wallis) rejects H_0, to find which pairs of groups differ. k = number of pairs.

✍️ **Best practice:** …

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Reading Python ANOVA tables (`smf.ols('y ~ factor')` + `sm.stats.anova_lm`)

- Columns: df, sum_sq (SSTR / SSE), mean_sq (sum_sq / df), F, PR(>F) = the **p-value**. Each row = one test; compare PR(>F) with α.
- Example one-way (income ~ Segment): F = 114.65, p ≈ 0 → reject H_0: mean income differs between segments.
- Example two-way (income ~ Segment*own_home): interaction p = 0.85 > 0.05 → no interaction, so the main effects can be read. Segment p ≈ 0 → significant; own_home p = 0.63 → not significant.
- Interaction plot: roughly parallel lines = no interaction; crossing or strongly diverging lines = interaction.

Reading reported results and Python

Report	What it tells you
t(50) = 8.2, p < .005	t-test with df 50: one-sample or paired → n = 51; two-sample → $n_1 + n_2 = 52$. Reject H_0.

Report	What it tells you
$F(3, 96) = 4.5, p = .005$	ANOVA: 3 = I − 1 → 4 groups; 96 = N − I → N = 100. Reject H_0 at $\alpha = 0.05$.
$\chi^2(1) = 15.98, p < .001$	chi-square with df 1 → a 2×2 table (independence) or 2 categories (goodness of fit).
$r = -.85, p < .01$	very strong negative relationship, significant.
$z = -2.67, p = .0038$	z-test: $\bar{x}$ lies 2.67 standard errors below μ_0.

- "$p < .005$" in an MC question: H_0 is rejected, and the risk that this is a Type I error is below .005. A Type II error can't apply (you rejected).
- With two groups, ANOVA gives the same result as the two-sample t-test ($F = t^2$).

Python basics

Code	Meaning	Example
/ and //	divide, whole-number divide	7 / 2 → 3.5, 7 // 2 → 3
%	remainder (modulo)	7 % 3 → 1, 3 % 3 → 0
**	power	2 ** 3 → 8
= and ==	assign, compare	i = i + 1; x % 2 == 0 → x is even
range(n)	0, 1, …, n − 1 (n not included)	range(3) → 0, 1, 2
break / continue	leave the loop / skip to the next round	

- Trace code line by line in a small table (round, variable values, what prints). The order of the lines inside the loop decides the output.
- ⚠️ range(5) stops at 4. Lists count from 0: in [10, 20, 30], position 0 is 10.

Tables

Z-table (standard normal, area to the LEFT of z)

Row = z to one decimal, column = the second decimal. Example: z = 1.53 → row 1.5, column .03 → 0.9370. Negative z (left half):

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Positive z (right half):

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Student's t table (Scribbr layout, also 1.1)

Row = df, column = α. Each value is the boundary t. Use the one-tailed α for < or > in H_a, and the two-tailed α for ≠ and confidence intervals (95% CI → two-tailed .05). df not in the table (e.g. 46)? Round down (40). The last row (∞) is the same as the z values.

df	1-tail 0.25 · 2-tail 0.5	1-tail 0.2 · 2-tail 0.4	1-tail 0.15 · 2-tail 0.3	1-tail 0.1 · 2-tail 0.2	1-tail 0.05 · 2-tail 0.1	1-tail 0.025 · 2-tail 0.05	1-tail 0.01 · 2-tail 0.02	1-tail 0.005 · 2-tail 0.01	1-tail 0.001 · 2-tail 0.002	1-tail 0.0005 · 2-tail 0.001
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646

df	1-tail 0.25 · 2-tail 0.5	1-tail 0.2 · 2-tail 0.4	1-tail 0.15 · 2-tail 0.3	1-tail 0.1 · 2-tail 0.2	1-tail 0.05 · 2-tail 0.1	1-tail 0.025 · 2-tail 0.05	1-tail 0.01 · 2-tail 0.02	1-tail 0.005 · 2-tail 0.01	1-tail 0.001 · 2-tail 0.002	1-tail 0.0005 · 2-tail 0.001
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
∞ (= z)	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
Confidence level	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%

The χ^2 table is under 5.1.

F table, $\alpha = 0.05$ (chapter 6 ANOVA)

Column = DF1 = I − 1 (numerator, SSTR). Row = DF2 = I(J − 1) (denominator, SSE). Example: DF1 2, DF2 18 → 3.555. DF not in the table: round down.

DF2 (row) / DF1 (column)	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385	19.396	19.413	19.429	19.446	19.454	19.462	19.471
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786	8.745	8.703	8.660	8.639	8.617	8.594
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.774	5.746	5.717
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.527	4.496	4.464
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.841	3.808	3.774
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.410	3.376	3.340
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.115	3.079	3.043
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.900	2.864	2.826
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.737	2.700	2.661
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.609	2.570	2.531
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.505	2.466	2.426
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.420	2.380	2.339
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.349	2.308	2.266
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.288	2.247	2.204
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.235	2.194	2.151
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.190	2.148	2.104
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.150	2.107	2.063
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.114	2.071	2.026
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.082	2.039	1.994
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.054	2.010	1.965
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	2.028	1.984	1.938
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	2.005	1.961	1.914
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.984	1.939	1.892
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.236	2.165	2.089	2.007	1.964	1.919	1.872
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.265	2.220	2.148	2.072	1.990	1.946	1.901	1.853
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204	2.132	2.056	1.974	1.930	1.884	1.836
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190	2.118	2.041	1.959	1.915	1.869	1.820
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177	2.104	2.027	1.945	1.901	1.854	1.806
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.887	1.841	1.792
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.793	1.744	1.693
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.700	1.649	1.594
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.608	1.554	1.495
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.517	1.459	1.394